Archives of Oral Biology 121 (2021) 104986

Contents lists available at ScienceDirect

Archives of Oral Biology

journal homepage: www.elsevier.com/locate/archoralbio

Comparative analysis of oral microbiome from Zang and Han populations living at different altitudes

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ARTICLE INFO

Keywords:
Oral microbiome
Diversity
Altitude
16S rRNA

ABSTRACT

Objectives: The aim of this study was to obtain greater insight into the environmental and genetic factors affecting the oral microbiome.
Design: To this end, we investigated the oral microbiome composition in Han and Zang populations living at different altitudes. The saliva microbiome in 115 individuals from Zang and Han populations living at different altitudes was analyzed using the 16 s rRNA gene sequencing method on the Illumina MiSeq platform. The dominant species in the oral microbiome were verified by quantitative real-time polymerase chain reaction (qPCR) analysis.
Results: The Han population, living at an altitude of 500 m, had higher microbiome diversity than the Zang population living at altitudes of 3000–4000 m. People living at 3000 m had a higher relative abundance of *Leptothrix* genus, but people living at 500 m had a higher relative abundance of *Capnocytophaga* genus according to Lefse difference analysis ($P < 0.05$). Compared to the Zang population, the Han population had higher relative abundances of *Porphyromonas* and *Treponema* genus organisms, especially *Porphyromonas* ($P < 0.001$). qPCR analysis confirmed that people living at high altitudes had the highest relative abundance of *Porphyromonas gingivalis* ($P < 0.01$).
Conclusions: This study showed that both genetics and the environment had significant influences on the oral microbiome composition. The study proposed a meaningful research direction to explore the relationship between different ethnic and altitude groups and oral diseases, such as periodontal diseases.

1. Introduction

The oral cavity is an important habitat for the microbiome, second only to the gastrointestinal tract (Chen, Yu, & Izard, 2010; Grice & Segre, 2012). More than 700 prokaryotic species exist in the oral cavity, 35 % of which have not been cultured, and there are 482 taxa with whole-genome sequences (Verma, Garg, & Dubey, 2018; Wade, 2013; Huttenhower, Gevers, & Knight, 2012). Advances in high-throughput sequencing technology have shifted the study of the oral microbiome from a single species to a community study, which has led to the discovery of a "core microbiome" (Gupta, Paul, & Dutta, 2017; Xiao, Ran, & Huang, 2016; Teng, Yang, & Huang, 2015). A number of studies have confirmed that an ecological imbalance in the oral microbiome can induce a variety of oral diseases, such as caries (Johansson, Witkowska, & Kaveh, 2016; Ma, Chen, & Zhang, 2015), apical periodontitis (Bouillaguet, Manoil, & Girard, 2018; Qian, Ma, & Ye, 2019), periodontal diseases (Acharya, Chen, & Chan, 2019; Eri Ikeda & Yuichi, 2019), oral mucosal disease (Carvalho, Cavalieri, & Do Nascimento, 2019; Wang, Lu, & Tu, 2016), and oral cancer (Chattopadhyay, Verma, & Panda, 2019; Hashimoto, Shimizu, & Hirabayashi, 2019; Yang, Yeh, & Yu, 2018). An ecological imbalance in the oral microbiome was also reported in systemic diseases, such as diabetes (Ogawa, Honda-Ogawa, & Ikebe, 2017; Sabharwal, Ganley, & Miecznikowski, 2019; Saeb, Al-Rubeaan, & Aldosary, 2019; Wang, Xu, & Ji, 2019; Wang, Rao, & Guo, 2019), cardiovascular diseases (Hansen, Egeberg, & Holmstrup, 2016; J-L, Stevens, & Paster, 2017; Yao, Akinrinade, & Chaix, 2020),

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https://doi.org/10.1016/j.archoralbio.2020.104986
Received 5 June 2020; Received in revised form 3 November 2020; Accepted 4 November 2020
Available online 12 November 2020
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autoimmune diseases (Dalmady, Kemeny, & Antal, 2020; Pessoa, Aleti, & Choudhury, 2019; Xuan, Zhang, & Jia, 2015), neurological diseases (Liu, Jiao, & Liao, 2019; Dragos Mihaila & Sarah, 2019), digestive diseases (Katarzyna Szczeklik & Dorota, 2019; Wang, Xu et al., 2019; Wang, Rao et al., 2019; Vogtmann, Han, & Caporaso, 2019), obesity (Ojcius, Tam, & Hoffmann, 2018; Yang, Cai, & Zheng, 2019; Wu, Chi, & Zhang, 2018), and preterm birth (Gomez-Arango, Barrett, & McIntyre, 2017; Pozo, Mesa, & Ikram, 2016). In view of the close relationship between the oral microbiome and health, it is crucial to determine the factors driving an ecological imbalance in the oral microbiome.

The factors influencing the oral microbiome include age, ethnicity, diet, oral hygiene, and the living environment and cultural background, which can be classified into genetic and external factors (Kato, Vasquez, & Moyerbrailean, 2017; Mason, Nagaraja, & Camerlengo, 2013; Fiorillo, 2020). Whether external factors and genetic factors have a primary or secondary influence on the oral microbiome is still unknown.

Previous studies showed that healthy individuals from different ethnic groups had differences in the structure and composition of their oral microbiome (Jing Li & Hans-Peter, 2014; Kim, Kang, & Lee, 2009; Mason et al., 2013; Ozga, Sankaranarayanan, & Tito, 2016). Some studies demonstrated that not only the oral microbiome structure and composition of Chinese minorities were different (Beibei, Li, & Yang, 2019; Kangli Jiao & Peiyuan, 2015), but also the prevalence of dental caries and periodontal diseases were different from those of the Han population (Fei Ma & Xueting, 2019; Luo, 2016; Mian, 2011). The Zang population lives in a typical high-altitude area, which has special environmental conditions, such as low barometric pressure, low temperature, low humidity, and high radiation. In addition, the special diet and insufficient awareness of oral hygiene have led to specific oral disease epidemics in the Zang population. Based on these characteristics of the Zang population, a few studies have discussed the relationship between the intestinal microbiome and the living altitudes of the Zang and Han populations (Li & Zhao, 2015; Li, Dan, & Gesang, 2016). The intestinal flora of the Zang population changed significantly with increasing altitude, the body mass index, and age, which might play an important role in regulating high altitude and geographical adaptability (Lan, Ji, & Lin, 2017). Zhilong et al. suggested that the general characteristics of the intestinal microbiota and clinical indexes of the Han population became increasingly similar to those of the Zang population with prolongation of plateau living (Jia, Zhao, & Liu, 2020). In terms of the oral microbiome, Zhiyan et al. indicated that the salivary microbiome diversity and abundance of the Han population were significantly higher than those of the Zang population, with high caries in Qinghai (Zhiyan & Jing, 2017). Some studies showed that the Zang population had a high incidence of dental caries and a high prevalence of periodontal diseases, which were aggravated with altitude and positively correlated with residence time on the plateau (Dongquan, Liu, & Song, 2009; Du, 2011).

However, a study on the structure and differences in the oral microbial communities in healthy Zang and Han populations living at different altitudes is still lacking. This research aimed to study the influence of different altitudes and ethnicities on the oral microbiome.

2. Materials and methods

2.1. Ethics statement

The experimental protocol was based on the ethical guidelines of the Declaration of Helsinki and approved by the Ethics Committee. Written informed consent was obtained from the individual participants. The privacy rights of the participants were always protected.

2.2. Subjects

The inclusion criteria were members of the Zang and Han population who had lived in Tibet for more than two years; members of the Zang and Han population who had lived in Chengdu for more than two years; those aged 18–80 years old; with at least 20 natural teeth; fewer or equal to six decayed, missing, or filled teeth (DMFT); and those who provided informed consent.

The exclusion criteria were the use antibiotics within three months; a history of dental treatment within six months; serious oral disease (periodontal diseases, oral mucosal disease, and oral cancer), and systemic diseases; pregnant or nursing women; dietary changes, and smoking more than 100 cigarettes in a lifetime (Yu, Phillips, & Gail, 2017).

2.3. Saliva sampling

As previous studies (Lan et al., 2017; Li & Zhao, 2015; Li et al., 2016), 30 subjects were recruited per group. According to the standard procedure, subjects were recruited in Hospital of Chengdu Office of People's Government of Tibetan Autonomous Region, which was based on the ethical guidelines of the Declaration of Helsinki and approved by the Ethics Committee. In this study, unstimulated saliva from 30 individuals of the Zang population living at altitudes of 3000–4000 m (group Zang in high altitude), 28 individuals of the Han population living in at altitudes of 3000–4000 m (group Han in high altitude), 27 individuals of the Zang population living at an altitude of 500 m (group Zang in low altitude), and 30 individuals of the Han population living at an altitude of 500 m (group Han in low altitude) was collected. Patients who signed informed consent were advised not to eat, drink, smoke, or chew gum 30 min before sampling. Before saliva collection, food residue in the mouth was rinsed off with drinking water, and at least 5 mL of saliva in the non-stimulated state was collected with a sterile centrifuge tube. After centrifugation, the saliva was stored in a 2 mL cryogenic tube and frozen at −80 °C immediately.

2.4. DNA extraction and Illumina MiSeq sequencing

The saliva samples were sent to Shanghai Majorbio Bio-pharm Technology Co., Ltd (Shanghai, China), where the microbial DNA was extracted, amplified, and sequenced according to their standard procedures. The detailed experimental procedure was performed according to a previous study (Wu, Ma, & Peng, 2020). The target V4–V5 regions of the 16S rRNA gene were amplified via polymerase chain reaction (PCR) using barcoded primers 515 F (5′-GTGCCAGCMGCCGCGG-3′) and 907R (5′−CCGTCAATTCMTTTRAGTTT-3′). The purified amplicons were pooled in equimolar amounts and paired-end sequenced (2 * 300) on an Illumina MiSeq platform (Illumina, San Diego, CA, USA) by Majorbio Bio-Pharm Technology Co., Ltd. according to standard protocols. The raw reads were deposited into the NCBI Sequence Read Archive (SRA) database (Accession Number: PRJNA613363).

2.5. Real-time quantitative PCR

Microbial DNA was extracted from the saliva samples. The final DNA concentration and purification were determined with a NanoDrop 2000 UV–vis spectrophotometer (Thermo Scientific, Wilmington, DE, USA), and DNA quality was checked by 1% agarose gel electrophoresis (Wu et al., 2020). The primers 5′- TAGCTTGCTAAGGTCGATGG-3′ and 5′-CAAGTGTATGCGGTTTTAGT-3′ were used to amplify *P. gingivalis* using a Light Cycler 480 qPCR machine (Roche, Basel, Switzerland) (Khemwong, Kobayashi, & Ikeda, 2019). Bacterial 16S rRNA universal primers 27 F (5′- AGAGTTTGATCMTGGCTCAG-3′) and 1429R (5′-CGGTTACCTTGTTACGACTT-3′) were used for the reference genes. The PCR reactions were conducted using the following program: 1 min of denaturation at 95 °C, 45 cycles of 10 s at 95 °C, 5 s of annealing at 55 °C, 10 s of elongation at 72 °C, and a 10 min of final extension at 72 °C. The PCR reactions were performed in triplicate in 20 μL mixtures containing 10 u L of SYBR Green, 6.4 μL of diethyl pyrocarbonate water, 2 μL of template DNA, and 0.8 u L of forward and reverse primers(10 u M) (< 100 ng).

2.6. Processing of sequencing data

Raw FASTQ files were quality-filtered by Trimmomatic and merged by FLASH with the standard criteria of Majorbio Bio-pharm Technology Co., Ltd. Operational taxonomic units (OTUs) were clustered with a 97 % similarity cutoff using UPARSE(version 7.1 http://drive5.com/uparse/) with a novel 'greedy' algorithm that performs chimera filtering and OTU clustering simultaneously. The taxonomy of each 16S rRNA gene sequence was analyzed by the Ribosomal Database Project Classifier algorithm (http://rdp.cme.msu.edu/) against the Silva database using a confidence threshold of 70 % (Cole et al., 2005; Dewhirst et al., 2010).

Mothur (version.1.30.2.) software was used to calculate alpha diversity, beta diversity, and species composition based on the OTU level. Principal component analysis (PCA) was performed according to the unweighted Unifrac distance matrix. Linear discriminant analysis effect size (LEfSe) was used for multilevel species discrimination analysis and linear discriminant analysis (LDA) was performed to detect the differences in microbiome composition between the groups. An LDA value greater than 3 was considered statistically significant. SPSS 21.0 software was used to analyze the experimental results by the Kruskal-Wallis test. A P-value of < 0.05 was considered statistically significant.

Fluorescent signals were collected in the qPCR analysis, the Ct (threshold) cycles of the gene and reference genes were determined, and the 2-Δ Δ CT values were calculated. The 2-Δ ΔCT values and the DNA concentration in each sample were analyzed using SPSS 21.0 software. One-way ANOVA and Duncan's least significant difference (LSD) test was used to compare the differences between the groups. A P-value of $<$ 0.05 was considered statistically significant.

3. Results

3.1. Taxonomic analysis

The Venn diagram of species composition showed that 852 OTUs were detected in group Han in low altitude, 798 OTUs were detected in group Zang in low altitude, 615 OTUs in group Zang in high altitude, and 893 OTUs in group Han in high altitude (Fig. 1a). The number of OTUs detected in the Han population was greater than that in the Zang population, and the group Han in high altitude had the maximum number of OTUs. There were 456 common OTUs in the four groups, accounting for 51.06%–74.15% of the total microbiome. There were 97 OTUs in group Han in low altitude, 77 OTUs in group Zang in low altitude, 30 OTUs in group Zang in high altitude, and 138 OTUs in group Han in high altitude. From the above data, it can be inferred that the oral microbiome of the four groups was relatively similar. The Han population generally had more abundant species composition, especially the group Han in high altitude (Fig. 1a).

The samples were divided into three groups according to the residence altitude (500 m, 3000 m, and 4000 m). There were 1023 OTUs in the group living at 500 m, 861 OTUs in the group living at 3000 m, and 708 OTUs in the group living at 4000 m. With a rise in altitude, the OTU number decreased. There were 236, 129, 30 OTUs found in the 500 m group, 3000 m group, and 4000 m group, respectively. There were 563 common OTUs in the three groups, accounting for 55.03%–79.52% of the total microbiome. To summarize, species diversity decreased with altitude increases, and the 500 m group had the most abundant species composition (Fig. 1b).

3.2. Alpha diversity analysis

The Ace index was used to describe the diversity of the oral

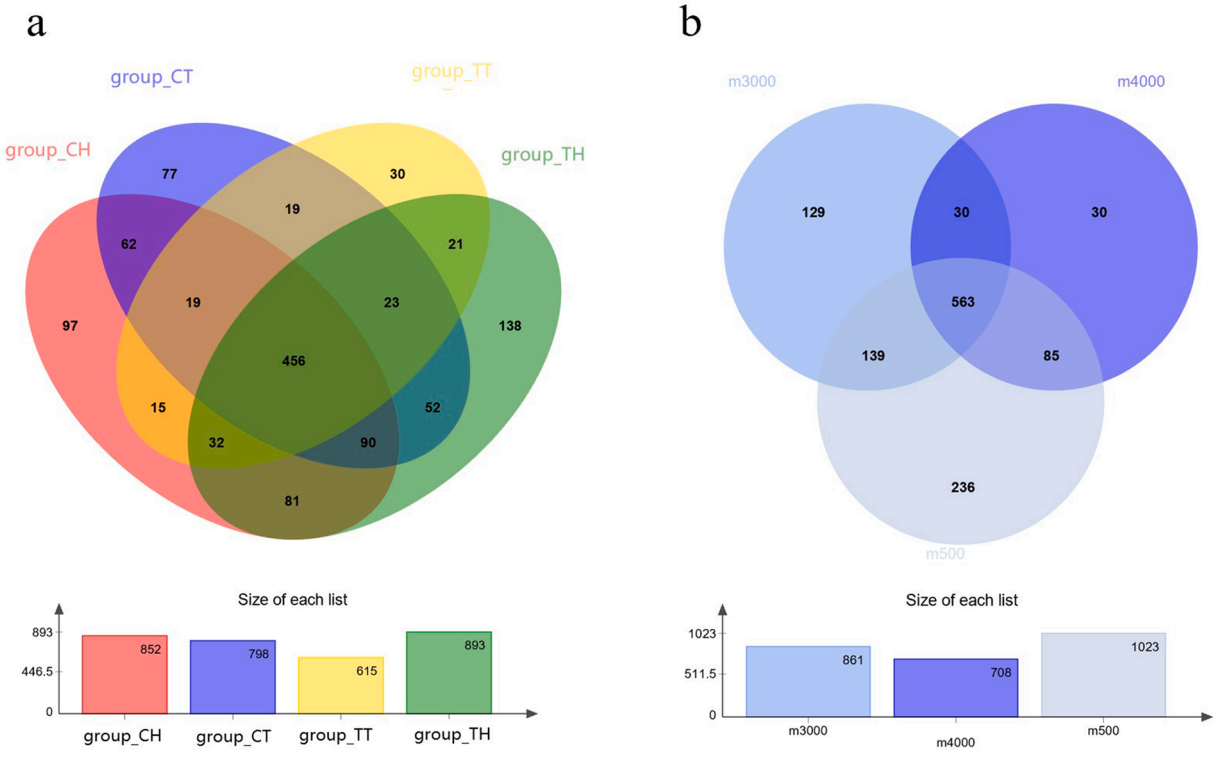


Fig. 1. Venn diagram of the species composition showed that: (a) 852 OTUs were detected in the group CH, 798 OTUs in group CT, 615 OTUs in group TT, and 893 OTUs in group TH. There were 456 common OTUs in the four groups, accounting for 51.06 % – 74.15 % of the total microbiome. (b) 1023 OTUs were detected in the 500 m group, 861 OTUs in the 3000 m group, and 708 OTUs in the 4000 m group. There were 563 common OTUs in the three groups, accounting for 55.03 % – 79.52 % of the total microbiome. OTUs: Operational taxonomic units; group_TT: the Zang population living at altitudes of 3000 – 4000 m; group_TH: the Han population living in at altitudes of 3000 – 4000 m; group_CT: the Zang population living at an altitude of 500 m; group_CH: the Han population living at an altitude of 500 m.

microbiomes. As illustrated in Fig. 2 (a), the Ace index in group Han in low altitude was significantly higher than that in the other groups ($P < 0.05$). The results indicated that the species richness of group Han in low altitude was significantly higher than in the other groups. The collected samples were divided into two groups according to ethnic differences, group Zang and group Han. There was no significant difference between the two groups according to the Ace index (Fig. 2b). In addition, the Ace index of the 500 m, 3000 m, and 4000 m groups showed no statistically significant differences (Fig. 2c). The collected samples were also divided into two groups according to different regions, the Chengdu group and the Tibet group. There was no significant difference between the two groups according to the Ace index (Fig. 2d).

3.3. Beta diversity analysis

PCA based on the relative abundance of OTUs revealed no significant separation in microbiome composition between the different groups (Fig. 3). The results showed that these groups had similar compositions and distribution of organisms in the oral microbiome.

3.4. LDA effect size (LEfSe) analysis

To study the differences in the oral microbiome between the Zang and Han populations living at different altitudes, a Lefse difference analysis was performed. In the Han population, the relative abundance of *Bacteroidia*, *Bacteroidetes*, *Bacteroidales*, *Porphyromonas*, and *Porphyromonadaceae* was higher than that in the Zang population. The relative abundance of *Lautropia* and *Proteobacteria* was higher in the oral microbiome of the Zang population compared to that of the Han population (Fig. 4a).

The taxonomic abundances in the oral microbiome of the Zang and Han populations living at low altitude were compared by Lefse analysis. More *Porphyromonas*, *Porphyromonadaceae*, *Fusobacteria*, and *Fusobacteriales* were found in the oral microbiome of the Han population living at low altitude, and more *Pasteurellaceae*, *Pasteurellales*, *Haemophilus*, and *Veillonella* were found in the oral microbiome of the Zang population living at low altitude (Fig. 4b).

Similarly, the taxonomic abundances in the oral microbiome of the Zang and Han populations living at a high altitude were compared by Lefse analysis. Higher abundances of *Bacteroidia*, *Bacteroidetes*, and

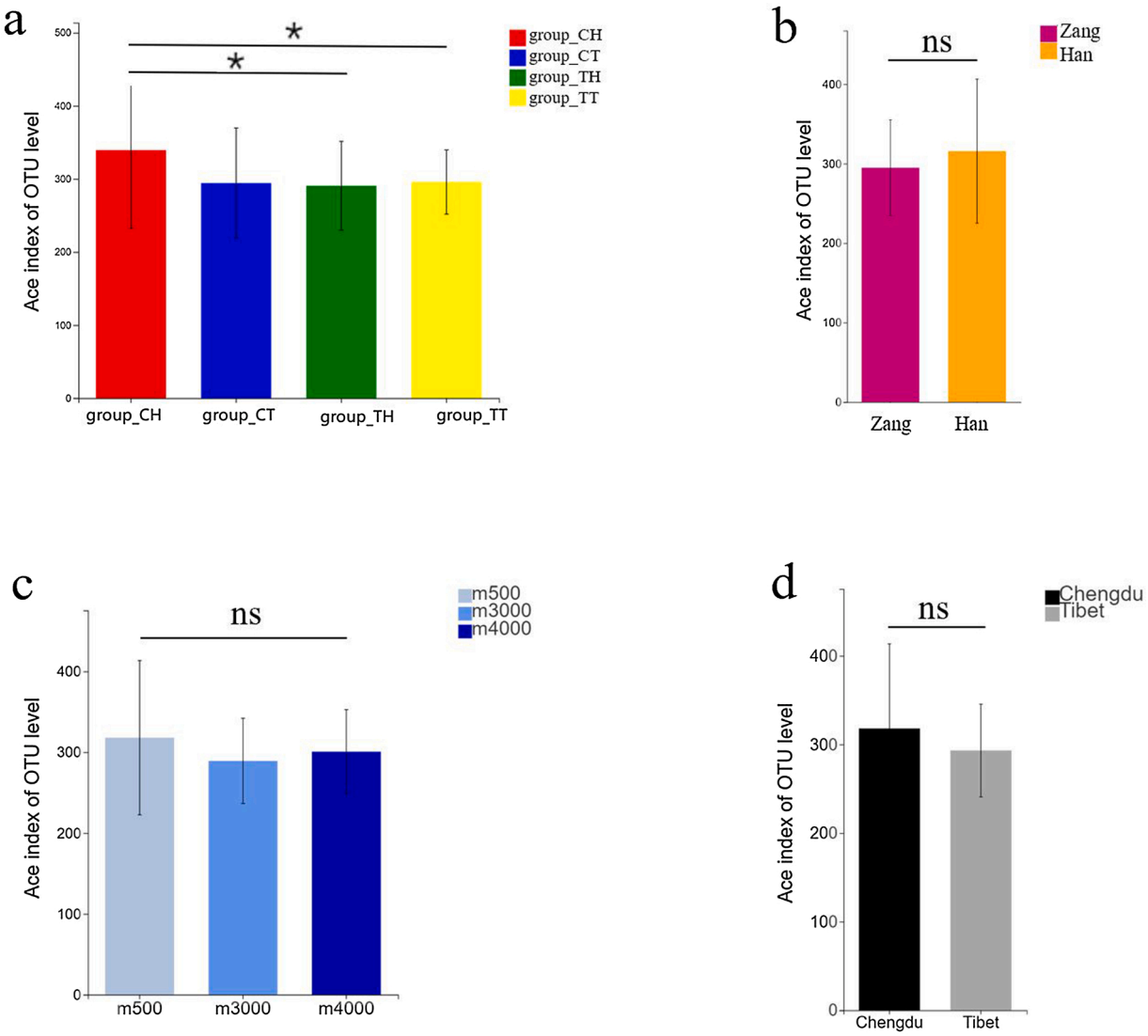


Fig. 2. Alpha diversity. (**a**) Ace indexes of groups CH, CT, TT, TH. The Ace index of group CH was significantly higher than that of the other groups. (**b**) Ace indexes of the Han and Zang groups. (**c**) Ace indexes of the 500 m, 3000 m, and 4000 m group. (**d**) Ace indexes of the Chengdu and Tibet groups. Group_TT: the Zang population living at altitudes of 3000 – 4000 m; group_TH: the Han population living in at altitudes of 3000 – 4000 m; group_CT: the Zang population living at an altitude of 500 m; group_CH: the Han population living at an altitude of 500 m. Data are presented as mean + SD. $*p < 0.05$.

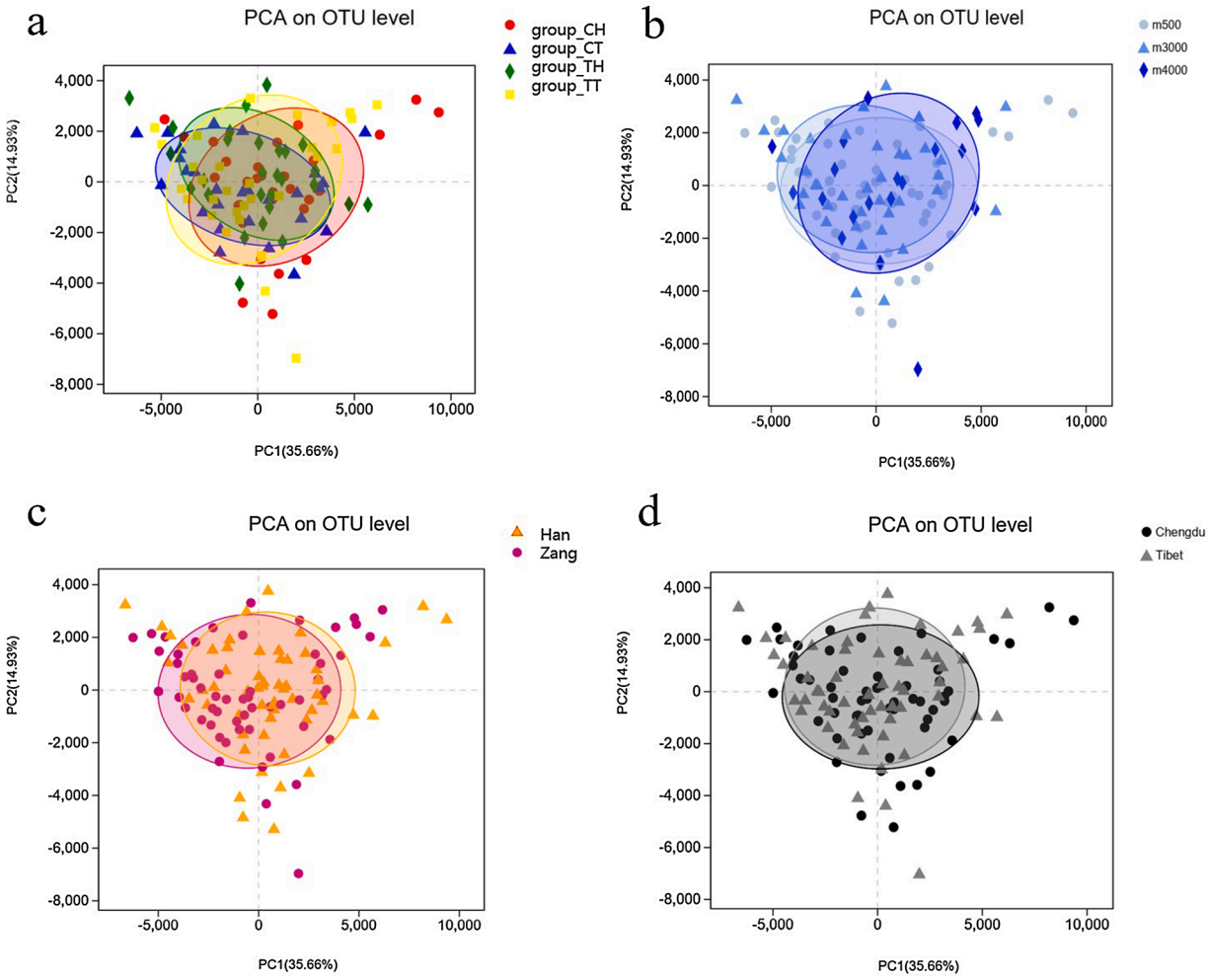


Fig. 3. PCA based on the relative abundance of OTU revealed no significant separation in microbiome composition between the different groups. (**a**) Groups CH, CT, TT, and TH; (**b**) 500 m, 3000 m, and 4000 m groups; (**c**) the Han and Zang groups; and (**d**) the Chengdu and Tibet groups. PCA: Principal component analysis; OTU: Operational taxonomic unit; group_TT: the Zang population living at altitudes of 3000 – 4000 m; group_TH: the Han population living in at altitudes of 3000 – 4000 m; group_CT: the Zang population living at an altitude of 500 m; group_CH: the Han population living at an altitude of 500 m.

Porphyromonadaceae were detected in the oral microbiome of the Han population living at a high altitude, and higher abundances of *Lautropia* and *Proteobacteria* were detected in the oral microbiome of the Zang population living at a high altitude (Fig. 4c).

The taxonomic abundances in the oral microbiome of the Han population living at low or high altitudes were compared by Lefse analysis. At low altitude, the Han population had higher abundances of *Capnocytophaga*, *Flavobacteriaceae*, *Flavobacteriales*, *Fusobacteriaceae,* and *Fusobacterium,* and a higher abundance of *Veillonellaceae* was found in the oral microbiome of the Han population living at a high altitude (Fig. 4d).

The taxonomic abundances in the oral microbiome of the Zang population living at low or high altitudes were compared by Lefse analysis. At low altitude, the Zang population had higher abundances of *Pasteurellaceae*, *Pasteurellales*, and *Haemophilus,* and greater proportions of *Fusobacteria* and *Fusobacteriales* were found in the oral microbiome of the Zang population living at a high altitude (Fig. 4e).

3.5. Analysis of differences in oral microbiome species composition

The differences in oral microbiome species composition of the groups Han in low altitude, Zang in low altitude, Zang in high altitude, Han in high altitude are shown in Fig. 5 (a). Groups Han in low altitude and Han in high altitude had a higher abundance of *Porphyromonas* than groups Zang in low altitude and Zang in high altitude ($P < 0.001$). The differences in oral microbiome species composition of the 500 m, 3000 m, and 4000 m groups are shown in Fig. 5 (b). The highest abundance of *Leptotrichia* was found in the 3000 m group, and the highest abundance of *Capnocytophaga* was found in the 500 m group ($P < 0.05$).

As illustrated in Fig. 5 (c), the Han population had a higher abundance of *Porphyromonas* ($P < 0.001$) and *Capnocytophaga* ($P < 0.05$). As illustrated in Fig. 5 (d), people living in Tibet had a higher abundance of *Leptothrix*, and people living in Chengdu had a higher abundance of *Capnocytophaga* ($P < 0.05$).

3.6. qPCR of P. gingivalis

The Circos plot of the sample and species collinearity of groups Han in low altitude, Zang in low altitude, Zang in high altitude, and Han in high altitude reflected the proportions of the dominant genus *Porphyromonas* in each sample (Fig. 6). The relative abundance of *P. gingivalis* was determined by qPCR from 10 saliva samples screened from each group. The results showed that the group Zang in high altitude had the highest abundance of *P. gingivalis* of all four groups ($P < 0.01$) (Fig. 7 a). A higher abundance of *P. gingivalis* was detected in the oral microbiome of people living in Tibet than in people living in Chengdu ($P < 0.05$) (Fig. 7 b), but there was no significant difference between the Zang and Han populations (Fig. 7 c).

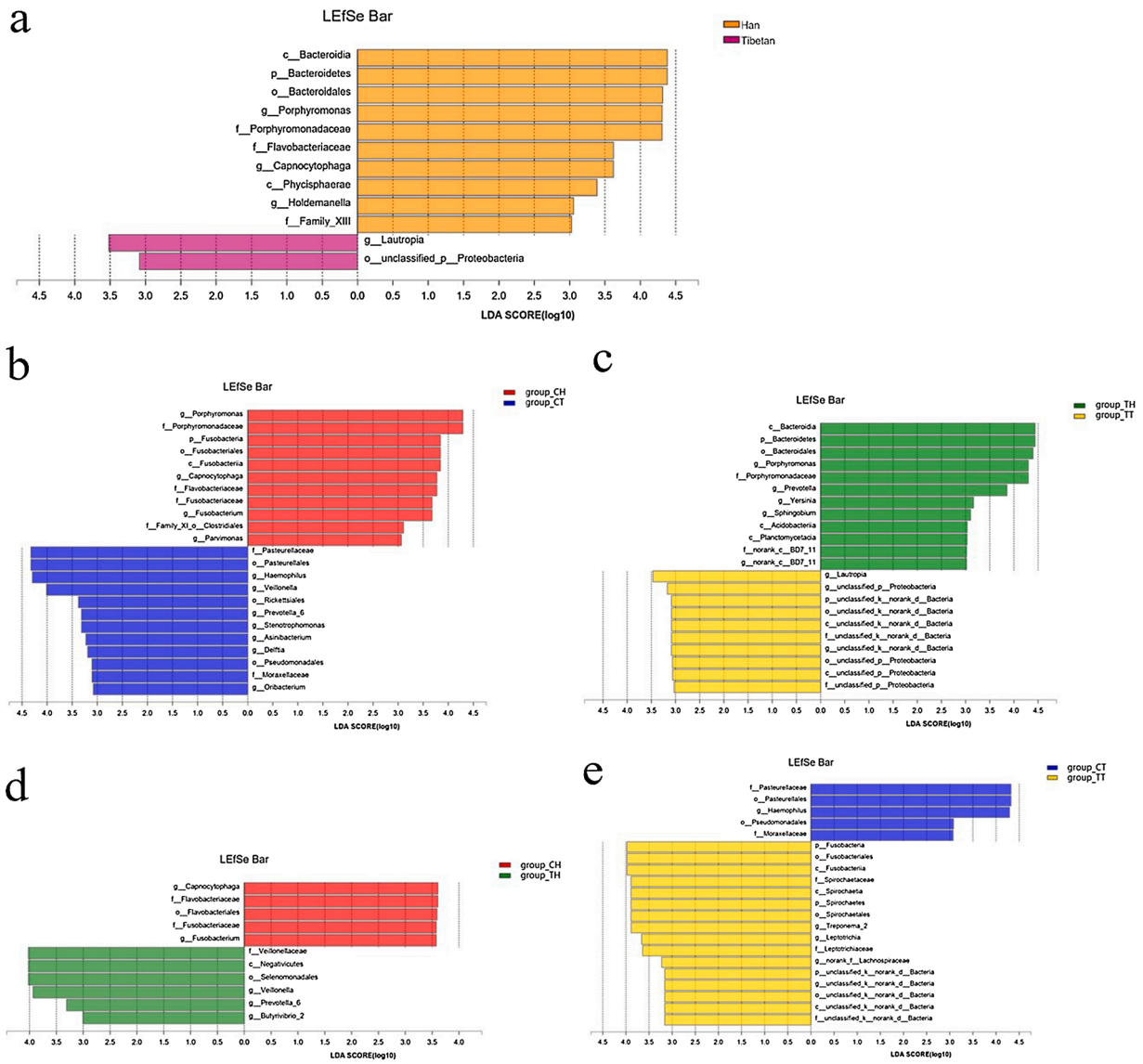


Fig. 4. LDA effect size analysis of differences in the microbiome between (**a**) the Han population and the Zang population. *Bacteroidia* ($p < 0.01$), *Bacteroidetes* ($p < 0.01$), *Bacteroidales* ($p < 0.05$), *Porphyromonas* ($p < 0.001$), *Porphyromonadaceae* ($p < 0.001$), *Lautropia* ($p < 0.05$), and *Proteobacteria* ($p < 0.05$); (**b**) groups CH and CT, Zang and Han populations living at low altitude (Chengdu): *Porphyromonas* ($p < 0.01$), *Porphyromonadaceae* ($p < 0.01$), *Fusobacteria, Fusobacteriales, Fusobacteriia* ($p < 0.05$), *Pasteurellaceae* ($p < 0.05$), *Pasteurellales* ($p < 0.05$), *Haemophilus* ($p < 0.05$), and *Veillonella* ($p < 0.05$); (**c**) groups TH and TT, Zang and Han populations living at a high altitude (Tibet): *Bacteroidia* ($p < 0.05$), *Bacteroidetes* ($p < 0.05$), *Bacteroidales* ($p < 0.05$), *Lautropia* ($p < 0.05$), and *Proteobacteria* ($p < 0.01$); (**d**) groups CH and TH, Han population living at low or high altitude (Chengdu or Tibet): *Capnocytophaga* ($p < 0.05$), *Flavobacteriaceae* ($p < 0.05$), *Flavobacteriales* ($p < 0.05$), *Fusobacteriaceae*, *Fusobacterium* ($p < 0.05$), and *Veillonellaceae* ($p < 0.05$); and (**e**) groups CT and TT, Zang population living at low or high altitude (Chengdu or Tibet): *Pasteurellaceae*, *Pasteurellales* ($p < 0.05$), *Haemophilus* ($p < 0.05$), *Fusobacteria*, *Fusobacteriia*, and *Fusobacteriales* ($p < 0.01$). LEfSe: Linear discriminant analysis effect size; group_TT: the Zang population living at altitudes of 3000 – 4000 m; group_TH: the Han population living in at altitudes of 3000 – 4000 m; group_CT: the Zang population living at an altitude of 500 m; group_CH: the Han population living at an altitude of 500 m. Data are presented as LDA_value.

4. Discussion

Our study revealed that people living in low-altitude Chengdu and the Han population had higher microbial community diversity and more abundant species composition. A lower abundance of *Porphyromonas* was found in the oral microbiome of the Zang population than the Han population ($P < 0.001$). However, the Zang population in Tibet had the highest abundance of *P. gingivalis*. To the best of our knowledge, this is the first report of the differences in the oral microbiome between Zang and Han populations living at different altitudes.

In recent years, rapid advances in sequencing methods and analytical techniques have enabled us to explore the human microbiome further (Moon, 2016). In our study, the 16 s rRNA sequencing method was used to analyze the microbiome at the genus level, while the dominant species were verified by qPCR. Social factors may be important potential confounders affecting the oral microbiome and health (Renson, Jones, & Beghini, 2019). Complex impact factors can be summarized as genetic or ethnic factors and environmental or acquired factors. The Zang population, one of the Chinese ethnic minorities, has maintained a special diet and high-altitude living environment for generations. Thus, the Zang and Han populations living at different altitudes were included to study the effects of different ethnicities and altitude on the oral microbiome composition.

A number of studies have discussed the effects of heredity and environment on the oral microbiome (Gupta et al., 2017; Mason et al., 2013; Kato et al., 2017; Zhiyan & Jing, 2017; Dongquan et al., 2009; Moon, 2016; Renson et al., 2019; He, Li, & Cao, 2015; Sun & Chang, 2014; Yang, Zheng, & Cai, 2019; Nasidze, Li, & Quinque, 2009). Mason

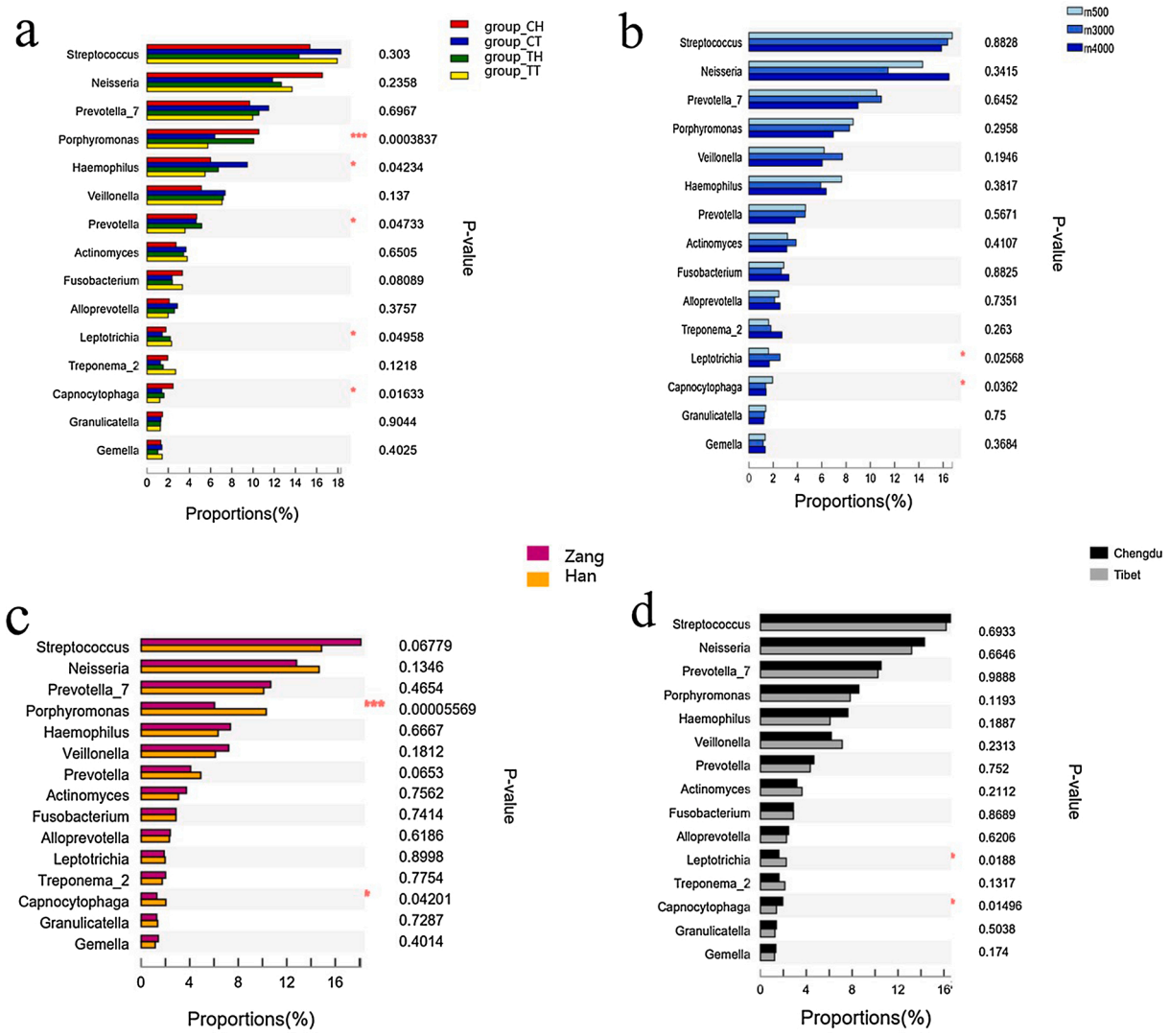


Fig. 5. Analysis of differences in oral microbiome species composition at the genus level. (**a**) Groups CH, CT, TT, and TH. Groups CH and TH had a higher abundance of *Porphyromonas*; (**b**) Altitude groups 500 m, 3000 m, and 4000 m. The 3000 m group had the highest abundance of *Leptotrichia*, and the 500 m group had the highest abundance of *Capnocytophaga*. (**c**) The Zang and Han populations. The Han population had a higher abundance of *Porphyromonas* and *Capnocytophaga*; (**d**) The group living in Chengdu and Tibet. People living in Tibet had a higher abundance of *Leptothrix*, and people living in Chengdu had a higher abundance of *Capnocytophaga*. Group_TT: the Zang population living at altitudes of 3000 – 4000 m; group_TH: the Han population living in at altitudes of 3000 – 4000 m; group_CT: the Zang population living at an altitude of 500 m; group_CH: the Han population living at an altitude of 500 m. Data are presented as mean. $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

et al. suggested that the oral subgingival microbiome of different ethnicities had obvious differences and suggested that there was a core microbial community among nationalities. They thought that the influence of ethnic and genetic factors was greater than the changes in environmental factors (Mason et al., 2013). A previous study reported significant differences among 69 OTUs and four bacterial genera (*Lactobacillus*, *Prevotella*, *Porphyromonas*, and *Fusobacterium*) due to sociodemographic characteristics (age, sex, income, education, birth status, and ethnicity) ($P < 0.01$) (Renson et al., 2019). As the results of LEfSe analysis revealed, in the Han population, the relative abundances of *Bacteroidetes*, *Porphyromonas*, and *Porphyromonadaceae* were higher than those in the Zang population. The relative abundances of *Lautropia* and *Proteobacteria* were higher in the oral microbiome of the Zang population compared to those of the Han population (Fig. 4a). The oral microbiome of the Han population had higher microbial community diversity and more abundant species composition. To some extent, these results were consistent with another relevant study in which the salivary microbiome diversity and abundance in the Han population with caries in Qinghai were significantly higher than those of the Zang population, and the dominant species were significantly different (Zhiyan Li & Jing, 2017). Except for ethnic differences, there were some contradictory significant differences in the oral microbiome between the Zang and Han populations, which may have been due to the influence of living altitude.

There are relatively few studies on the influence of altitude on the oral microbiome composition. Interestingly, *Bacteroidetes* was the dominant species among both the intestinal and oral microbiome of an immigrant Han population compared to the native Tibetan population (Li et al., 2016). The Zang population at low altitude had higher abundances of *Pasteurellaceae* and *Haemophilus* ($P < 0.05$) and a greater proportion of *Fusobacteria* was found in the oral microbiome of the Zang population at a high altitude ($P < 0.01$) (Fig. 4e). Given that the two Zang population groups basically shared the typical Tibetan lifestyle and dietary habits, it can be speculated that differences in the living environment caused by altitude led to the different compositions of the oral microbial community. The special plateau environment is characterized

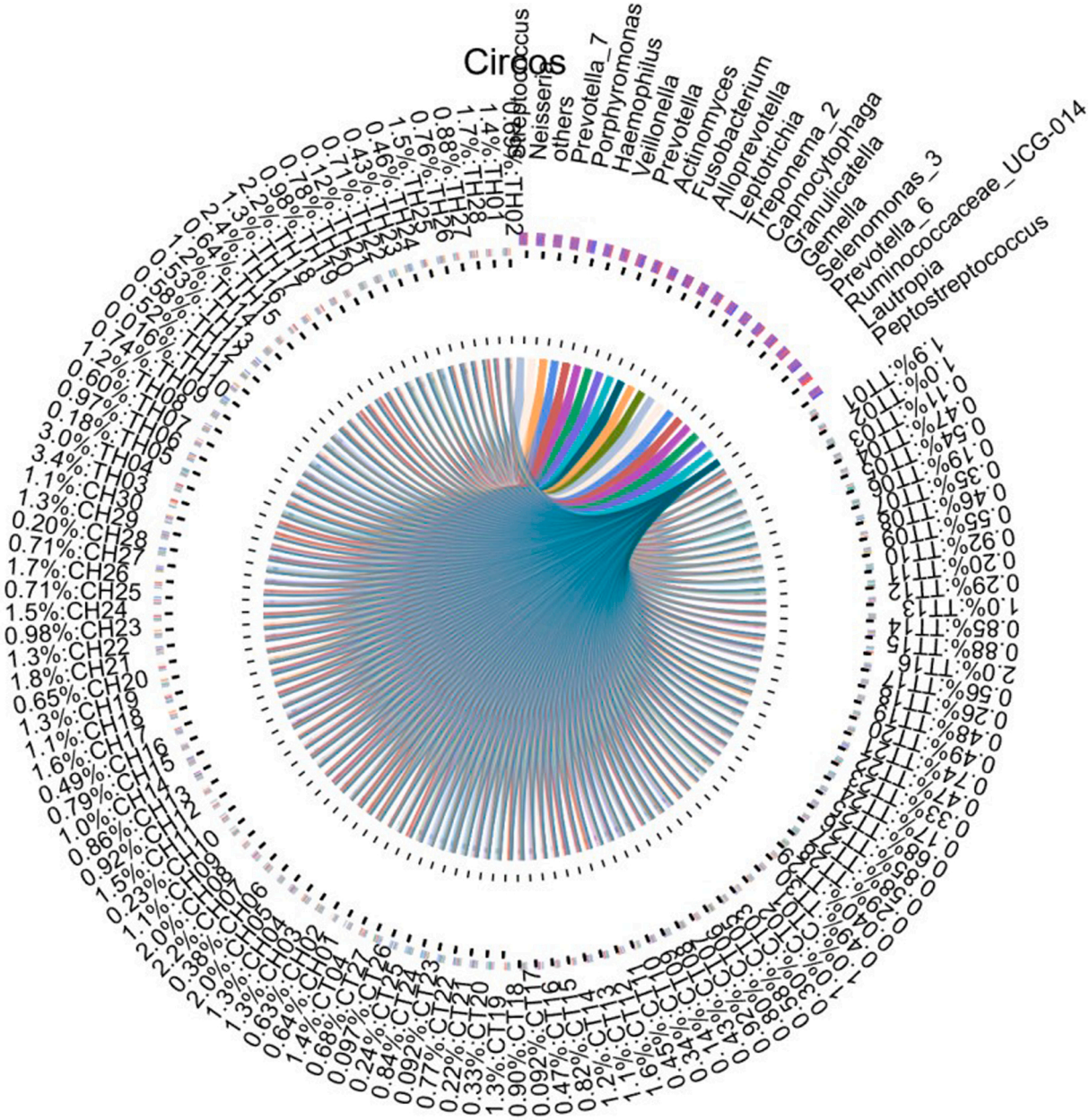


Fig. 6. Circos plot of sample and species collinearity of the CH, CT, TT, and TH groups. The small semicircle indicates the species composition in the sample. The greater half circle represents the proportion abundance of *Porphyromonas* distributed in each sample. TT: the Zang population living at altitudes of 3000 – 4000 m; TH: the Han population living in at altitudes of 3000 – 4000 m; CT: the Zang population living at an altitude of 500 m; CH: the Han population living at an altitude of 500 m.

by low air pressure, low oxygen, low temperature, low humidity, and high radiation due to its high altitude. Low oxygen is conducive to the dominant growth of anaerobic bacteria, and the microbial community diversity decreased compared to that in the low-altitude environment (Zhiyan Li & Jing, 2017; Dongquan et al., 2009; Jianzhen She & Xingtian, 2016).

The saliva microbiome compared by genus (Fig. 5) showed that a higher proportion of *Porphyromonas* was detected in groups Han in low altitude and Han in high altitude than in groups Zang in high altitude and Zang in low altitude ($P < 0.001$) and the Han population had a higher proportion of *Porphyromonas* than the Zang population ($P < 0.001$). When grouped by altitude, the highest abundance of *Leptotrichia* was found in the 3000 m group, and the highest abundance of *Capnocytophaga* was found in the 500 m group ($P < 0.05$). The results were consistent with people living in Chengdu who had a greater abundance of *Capnocytophaga*, and people living in Tibet who had a greater abundance of *Leptotrichia* ($P < 0.05$). *Leptotrichia* and *Capnocytophaga* were reported to be associated with periodontal diseases (Ge, Rodriguez, & Trinh, 2013; Scapoli, Girardi, & Palmieri, 2015; Scher, Ubeda, & Equinda, 2012). *Capnocytophaga* is a kind of oral opportunistic pathogen, which grows in an anaerobic environment with a high CO_2 concentration of 10 % (Felix, Rosenberg, & Caraballo, 2019). The Tibetan plateau has a thin atmosphere, and the CO_2 concentration is correspondingly low. Tibetan dietary habits are mainly composed of high-fat buttered tea, cheese, and meat. Murtaza et al. found that the relative abundance of *Haemophilus*, *Neisseria,* and *Prevotella* were decreased, and the relative abundance of *Streptococcus* was increased in the oral microbiome of elite endurance athletes following the consumption of a low carbohydrate and high-fat diet (Murtaza, Burke, & Vlahovich, 2019). Compared to the Han population, higher proportions of *Neisseria* and *Prevotella* and a

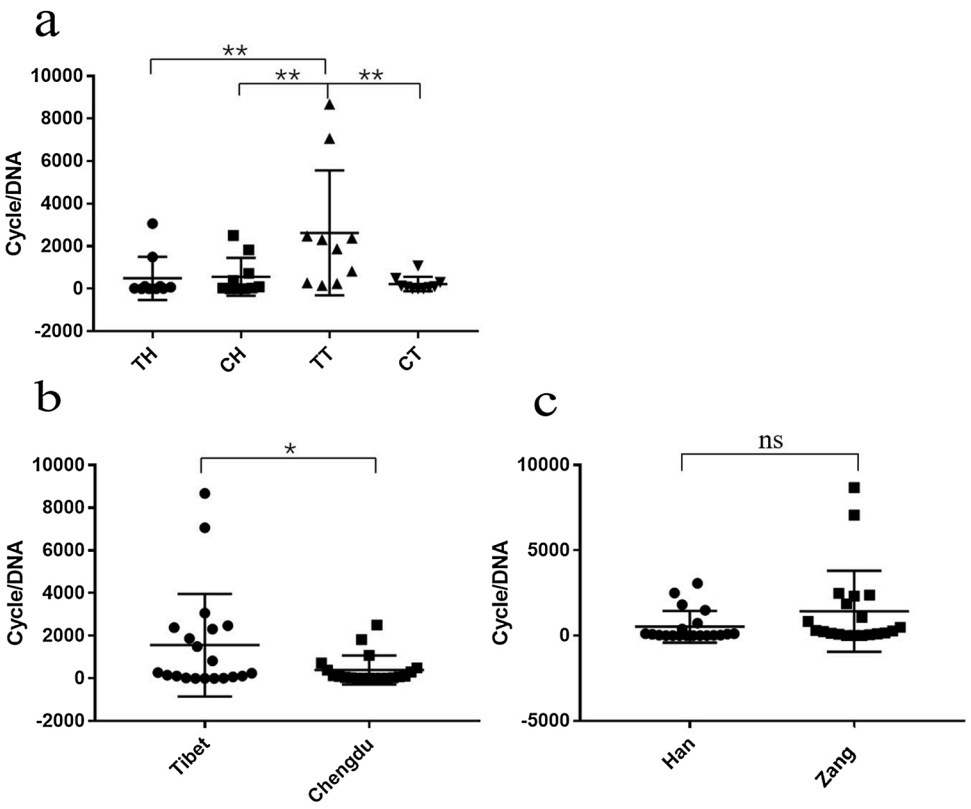


Fig. 7. The relative abundance of *P. gingivalis* was determined by qPCR from 10 saliva samples screened from each group. (**a**) Group CH, CT, TT, and TH. (**b**) Tibet and Chengdu groups. (**c**) The Zang and Han populations. TT: the Zang population living at altitudes of 3000 – 4000 m; TH: the Han population living in at altitudes of 3000 – 4000 m; CT: the Zang population living at an altitude of 500 m; CH: the Han population living at an altitude of 500 m. Data are presented as mean + SD. $*p < 0.05$, $**p < 0.01$.

lower proportion of *Streptococcus* were found in the oral microbiome of the Zang population (Fig. 5c).

Notably, a lower abundance of *Porphyromonas* was found in the oral microbiome of the Zang population compared to the Han population ($P < 0.001$) (Fig. 5c). However, the Zang population living in Tibet had the highest abundance of *P. gingivalis* ($P < 0.01$) (Fig. 7a), and people living at a high altitude in Tibet all had a higher abundance of *P. gingivalis* ($P < 0.05$) (Fig. 7b). *P. gingivalis* is frequently detected in saliva from periodontitis patients, which is closely related to the occurrence and development of periodontal diseases (Damgaard, Danielsen, & Enevold, 2019). In addition, previous studies indicated that *P. gingivalis* was related to oral squamous cell carcinoma (Liu et al., 2020) and system diseases, such as cognitive disorders, Alzheimer`s disease, type 2 diabetes mellitus, obesity, and cardiovascular diseases (Carter, Crean, & Singhrao, 2017). The subjects in this study had no obvious symptoms/signs of periodontitis, but the high abundance of *P. gingivalis* in the oral microbiome of people living at a high altitude explained the relatively high incidence of periodontal diseases among the people living at high altitudes (Dongquan et al., 2009; Du, 2011; Jianzhen She & Xingtian, 2016). Animal experiments revealed that hypoxia may induce severe periodontitis by affecting the microbial community in gingival crevicular fluid (Terrizzi, Fernandez-Solari, & Lee, 2013; Xiao, Li, & Zhang, 2012). In a low-oxygen environment, different degrees of hypoxia were produced in organs of the body. Saliva secretion decreased, and the oral self-cleaning function was reduced, which accelerated the growth of periodontal anaerobic bacteria and the formation of a microbiome community (Zhiyan Li & Jing, 2017; Dongquan et al., 2009; Yang, Zheng et al., 2019).

Major limitation of the current study was that sample size cannot be appropriately estimated without population parameters (mean, standard deviation, etc.). We referred to several similar studies and chose the sample size. Based the results of this study, further large-scale studies should take into account this problem. And it is essential to explore the effects and mechanisms of specific oral microbiome compositions on the incidence of oral disease.

5. Conclusions

In summary, our study revealed that people living in low-altitude Chengdu and the Han population had higher oral microbiome microbial community diversity and more abundant species composition. The Zang population in Tibet had the highest abundance of *P. gingivalis*, in which altitude played a crucial role. No clear mechanism was identified for the influence of environmental and genetic factors on the oral microbiome. This study initially revealed significant differences in the oral microbiome between the Zang and Han populations at different altitudes, which provides some insight for further studies on the relationship between different geographical and ethnic factors and oral microbiome dysbiosis-related diseases.

Declaration of Competing Interest

None.

Acknowledgments

This research was supported by the National Natural Science Foundation of China 81870759 (LC), the Youth Grant of the Science and Technology Department of Sichuan Province, China [2017JQ0028 (LC)] and Innovative Research Team Program of Sichuan Province (L.C).

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